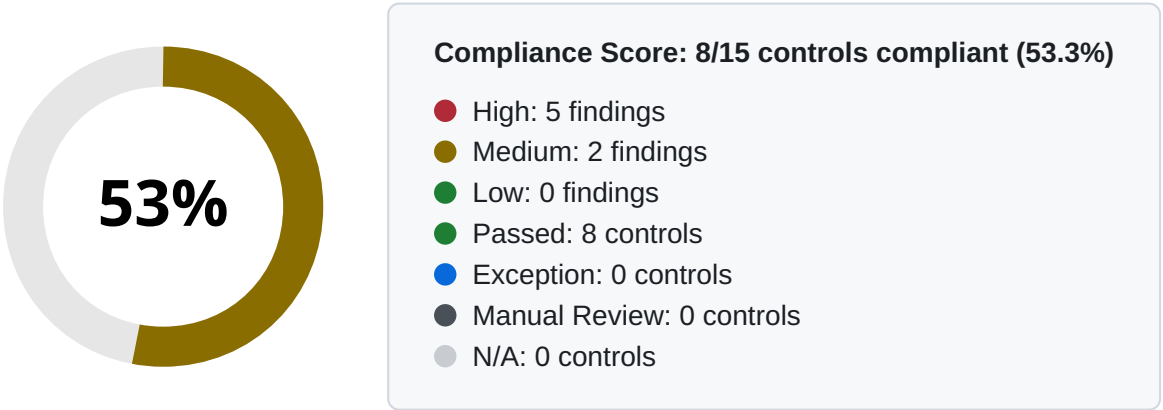


Cisco IOS-XE Compliance Report

CONFIDENTIAL - INTERNAL USE

Device:	device_config
Device role:	edge-router
Audit date:	2026-09-11
controls.yaml version:	Agent-Assisted Coding controls.yaml v9b0921f
Generated by:	Agent-Assisted Coding Compliance Report Generator

Executive Summary



METRIC	COUNT
Total controls evaluated	15
Compliant	8
Non-compliant (High)	5
Non-compliant (Medium)	2
Non-compliant (Low)	0
Exception granted	0
Manual review required	0
Not applicable	0

Scope & Methodology

This audit evaluated the device configuration listed above against the Agent-Assisted Coding controls framework (controls.yaml v9b0921f), covering the Agent-Assisted Coding 3-plane hardening model (management, control, and data plane) and the Cisco IOS-XE

hardening practices it encodes. Controls without a mapped external CIS Cisco IOS-XE Benchmark section are cited by their Agent-Assisted Coding control ID only; no CIS section reference is claimed unless a real mapping has been supplied for that control.

Deterministic controls were evaluated automatically against the device's running configuration. Controls marked "Manual Review" require human judgment (e.g. banner wording against corporate legal policy) and are listed separately below rather than auto-scored. Findings already covered by a documented exception are still listed in Findings Detail, marked accordingly, rather than silently excluded.

Out of scope: physical security, change-management process compliance, and any control outside the Agent-Assisted Coding framework's current 17 controls.

Findings Detail

[control_00004] TACACS — HIGH

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

Failure to communicate with central security servers forces the use of local accounts, making user management difficult and less secure across the network.

Evidence

'tacacs server ACME-TEAM-TACACS-01': no 'address ipv4' command found.; 'tacacs server ACME-TEAM-TACACS-01' uses a reversible Type 7 key; Type 6 encryption ('key config-key password-encrypt' + 'key 6 <key>') is required.; 'tacacs server ACME-TEAM-TACACS-02': no 'address ipv4' command found.; 'tacacs server ACME-TEAM-TACACS-02' uses a reversible Type 7 key; Type 6 encryption ('key config-key password-encrypt' + 'key 6 <key>') is required.; TACACS group name 'ACME-TEAM-TACACS' does not match the corporate group 'ACME_TACACS_SERVER_GROUP'.; No 'ip tacacs source-interface' configured under the TACACS server group.

Remediation Command

```
aaa new-model
! Repeat the following 'tacacs server' block for each TACACS+
server.
tacacs server <tacacs_server_name>
  address ipv4 <tacacs_server_address>
  key 6 <tacacs_key>
  timeout <tacacs_timeout>
aaa group server tacacs+ <tacacs_group_name>
  server name <tacacs_server_name>
  ip tacacs source-interface <source_interface>
```

Operator Inputs Required

<source_interface>, <tacacs_group_name>, <tacacs_key>,
<tacacs_server_address>, <tacacs_server_name>, <tacacs_timeout>

Owner

Network Security Team

Target Remediation Date

2026-09-26 (<= 15 days, or next maintenance window)

[control_00005] Local and emergency users — HIGH

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

Lack of local emergency accounts can result in total loss of access during a network or TACACS outage.

Evidence

Local account(s) use a weak password-hashing type: ['5'] (expected type 8/9 scrypt).;
Local accounts use inconsistent password-hashing types: ['5', '8'].

Remediation Command

```
username <user> privilege <privilege_level> algorithm-type  
scrypt secret <password>  
enable secret <secret_psswd>
```

Operator Inputs Required

<password>, <privilege_level>, <secret_psswd>, <user>

Owner

Network Security Team

Target Remediation Date

2026-09-26 (<= 15 days, or next maintenance window)

[control_00007] Global password encryption — HIGH

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

Without this command, TACACS+/RADIUS/routing keys fall back to weak Type-7 encoding or cleartext storage, trivially decodable if an attacker obtains the running-config, compromising network infrastructure.

Evidence

'key config-key password-encrypt <device_master_key>' is missing.; 'password encryption aes' is missing.

Remediation Command

```
key config-key password-encrypt <device_master_key>
password encryption aes
```

Operator Inputs Required

<device_master_key>

Owner

Network Security Team

Target Remediation Date

2026-09-26 (<= 15 days, or next maintenance window)

[control_00011] SNMP — HIGH

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

SNMPv1/v2c transmit community strings in clear text.

Evidence

SNMPv3 group with 'priv' is not configured.; SNMPv3 user is not created with SHA/AES configuration.; SNMP trap host is missing.; 'snmp-server enable traps' command is missing.

Remediation Command

```
snmp-server group <group_name> v3 priv
snmp-server user <user_name> <group_name> v3 auth sha
<auth_password> priv aes 128 <priv_password>
snmp-server host <snmp_server_ip> version 3 priv <user_name>
snmp-server enable traps
```

Operator Inputs Required

<auth_password>, <group_name>, <priv_password>, <snmp_server_ip>, <user_name>

Owner

Network Security Team

Target Remediation Date

2026-09-26 (<= 15 days, or next maintenance window)

[control_00014] VTY Lines — HIGH

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

Open VTY/Console sessions without timeouts can be hijacked.

Evidence

'line con 0': 'login local' is missing.; 'line con 0': 'exec-timeout' is missing.; 'line vty 0 4': 'login local' is missing.; ACL 'VTY-OPEN' bound to VTY lines is permissive instead of restrictive: permit tcp any any eq 22; permit tcp any any eq telnet; 'line vty 5 15': 'login local' is missing.

Remediation Command

```
line con 0
password <console_password>
login local
exec-timeout <minutes> <seconds>
logging synchronous
exit
line vty <start_line_number> <end_line_number>
access-class <vtty_acl_name> in
password <vtty_password>
login local
exec-timeout <minutes> <seconds>
transport input <allowed_protocols>
logging synchronous
exit
```

Operator Inputs Required

<allowed_protocols>, <console_password>, <end_line_number>, <minutes>, <seconds>, <start_line_number>, <vtty_acl_name>, <vtty_password>

Owner

Network Security Team

Target Remediation Date

2026-09-26 (<= 15 days, or next maintenance window)

[control_00009] NTP — MEDIUM

Status

NON-COMPLIANT

Reference

Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement

Drifting clocks lead to inaccurate log timestamps, making forensic analysis and event correlation during a security incident nearly impossible.

Evidence

'prefer' keyword is missing on the primary NTP server.; NTP authentication is not configured ('ntp authenticate' / 'ntp authentication-key' / 'ntp trusted-key' missing).

Remediation Command

```
ntp authentication-key <ntp_key_id> md5 <ntp_key_value>
ntp trusted-key <ntp_key_id>
ntp authenticate
ntp server <ntp_ip_address> prefer
ntp server <secondary_ntp_ip_address>
```

Operator Inputs Required

<ntp_ip_address>, <ntp_key_id>, <ntp_key_value>,
<secondary_ntp_ip_address>

Owner

Network Security Team

Target Remediation Date

2026-10-11 (<= 30 days)

[control_00010] Syslog — MEDIUM

Status
NON-COMPLIANT

Reference
Agent-Assisted Coding controls.yaml v9b0921f

Risk Statement
Without centralized logging, local logs may be overwritten or cleared by an attacker, leaving no trace of unauthorized activity.

Evidence
'logging on' command is missing.; Logging trap level is not set to 'informational'.; 'logging source-interface' is not configured.

Remediation Command

```
logging on
logging host <ip_syslog_server>
logging trap informational
logging source-interface <syslog_interface>
```

Operator Inputs Required
<ip_syslog_server>, <syslog_interface>

Owner
Network Security Team

Target Remediation Date
2026-10-11 (<= 30 days)

Manual Review Required

No controls require manual review on this audit.

Full Control Appendix

All 15 evaluated controls, regardless of outcome.

CONTROL	TITLE	SEVERITY	STATUS
control_00001	Hostname	Low	Compliant
control_00002	Domain	Low	Compliant
control_00003	AAA	High	Compliant
control_00004	TACACS	High	Non-Compliant
control_00005	Local and emergency users	High	Non-Compliant

CONTROL	TITLE	SEVERITY	STATUS
control_00006	SSH	High	Compliant
control_00007	Global password encryption	High	Non-Compliant
control_00008	ACL for VTY	High	Compliant
control_00009	NTP	Medium	Non-Compliant
control_00010	Syslog	Medium	Non-Compliant
control_00011	SNMP	High	Non-Compliant
control_00012	Banners	Medium	Compliant
control_00013	Passwords	High	Compliant
control_00014	VTY Lines	High	Non-Compliant
control_00015	Local user and enable secret	High	Compliant

Remediation Roadmap

SEVERITY	CONTROL	OWNER	TARGET REMEDIATION DATE
High	[control_00004] TACACS	Network Security Team	2026-09-26 (<= 15 days, or next maintenance window)
High	[control_00005] Local and emergency users	Network Security Team	2026-09-26 (<= 15 days, or next maintenance window)
High	[control_00007] Global password encryption	Network Security Team	2026-09-26 (<= 15 days, or next maintenance window)
High	[control_00011] SNMP	Network Security Team	2026-09-26 (<= 15 days, or next maintenance window)
High	[control_00014] VTY Lines	Network Security Team	2026-09-26 (<= 15 days, or next maintenance window)
Medium	[control_00009] NTP	Network Security Team	2026-10-11 (<= 30 days)
Medium	[control_00010] Syslog	Network Security Team	2026-10-11 (<= 30 days)

Sign-off

Network Team - Name / Signature	Role	Date
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Security Team - Name / Signature	Role	Date
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